

Smriti स्मृति

Personal Agentic RAG: fast and slow thinking for your studies and documents. Private, measurable and free.

“Handle your documents safely and securely.”



THE PROBLEM

Students' knowledge is scattered, and AI doesn't help safely



Scattered material

Notes, books, slides and past papers are spread across folders and phones



Hard to find

Marksheets, certificates and exam dates when you need them



Ungrounded answers

Chatbots answer from the internet, not your syllabus, and don't cite the page



Privacy cost

Cloud AI only works if your documents leave your control

Goal: an assistant that is **grounded**, **private** and **measurable**.

THE CORE IDEA

Think fast when you can, slow when you must

System 1 • Fast

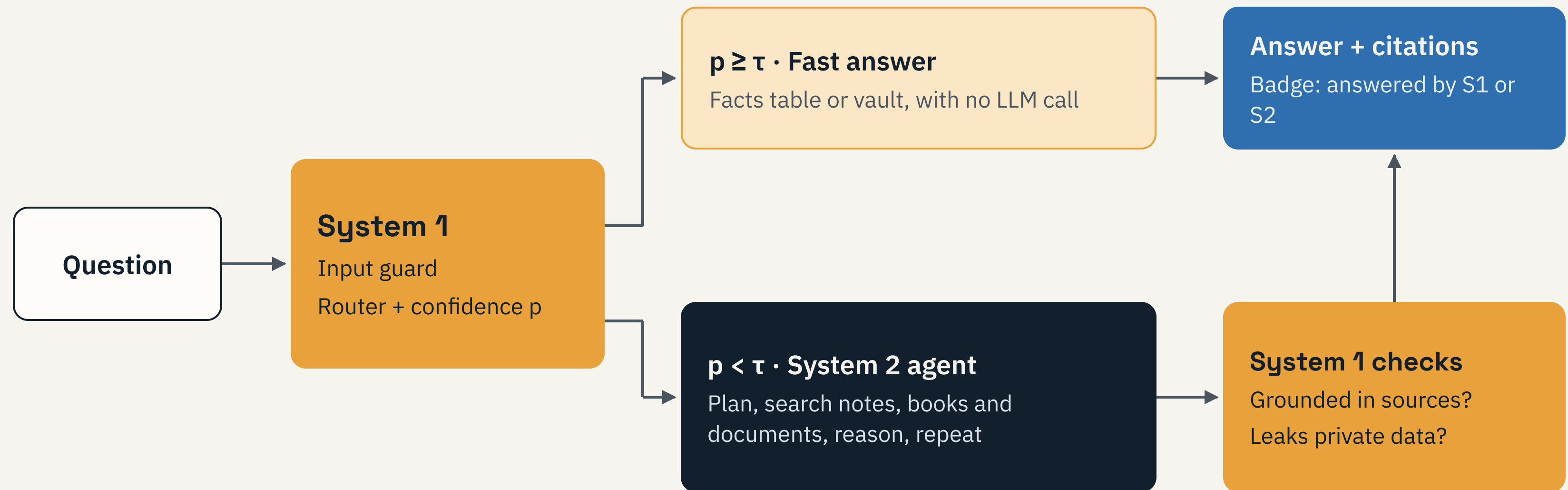
- Small model with typed answers + confidence
- Routes questions, answers simple facts
- Safety checks on every input and output
- No LLM call

System 2 • Slow

- LLM agent that plans and reasons
- Searches notes, books and documents
- Several steps, cites its sources
- Used only when System 1 is unsure

Inspired by Daniel Kahneman, *Thinking, Fast and Slow* (2011)

Escalate only when unsure



TWO EXAMPLES

Same assistant, two ways of thinking

FAST PATH · SYSTEM 1

“When is my DBMS exam?”

1. Guard: safe
2. Router: fact lookup, high confidence
3. Answer from the facts table
4. Cites the exam timetable PDF

No LLM call

SLOW PATH · SYSTEM 2

“Compare how my notes and Galvin explain paging”

1. Router: comparison, so escalate
2. Agent searches notes, then the book
3. Grounding check passes
4. Cites notes p. 14 and Galvin ch. 9

WHAT IT KNOWS

Your studies and your documents



Study library

- Lecture notes, books, slides
- Previous year question papers
- Organised: Semester → Course → Topic
- Answers cite the page number



Personal vault

- Marksheets, certificates, IDs
- Exam timetable and deadlines
- Encrypted, never sent to the cloud
- Facts extracted for fast answers

Built today: PDF, Markdown, TXT and DOCX with page numbers, plus automatic **PII redaction**. Next: **OCR** for scans and phone photos.

Three System-1 guards, plus a locked vault

G1 · INPUT

Injection guard

Blocks attacks in questions and in text hidden inside uploaded PDFs

G2 · OUTPUT

Grounding check

Is every claim supported by a cited source?
If not, retry or say “I don’t know”

G3 · OUTPUT

PII leak check

Never reveals ID numbers or marks you didn’t ask for



AES-256 encrypted vault



Local models by default



Audit log + red-team tests

Every decision is traced

Explain the four conditions for deadlock from my OS notes

Used Thinking · System 1 → System 2 · 9 steps ^

Step	Decision / detail	ms
s1.guard.input	safe (p_injection=0.02)	0
s1.router	explain (p=0.70)	0
escalate	to System 2 (intent=explain, p=0.70, tau=0.75)	0
s2.plan	Explain the four conditions for deadlock from my OS notes	0
s2.tool.search	'Explain the four conditions for deadlock from my OS notes' course=OS → 5 hits	2
s1.guard.chunks	stripped injected text from: shared_threads_notes p.1	0

Built into every answer

Each step with its decision, confidence and time, in the chat UI and the CLI (--trace)

Real run, shown left

Router escalates, the injected chunk is stripped, and grounding passes 2/2

OpenTelemetry ready

Optional export to Arize Phoenix; dashboards next

A working prototype, already open source

- ✓ System 1 rules engine: router, injection guard, grounding
- ✓ Fast answers from a facts table, with no LLM
- ✓ System 2 agent on a local LLM (Ollama)
- ✓ AES-256 vault, PII redaction, audit log
- ✓ Page-cited search over notes and books
- ✓ Chat UI, CLI, eval harness
- ✓ 41 tests and CI on GitHub

When is my DBMS exam?

Used Thinking · System 1 · 3 steps ^

Step	Decision / detail	ms
s1.guard.input	safe (p_injection=0.02)	0
s1.router	fact_lookup (p=0.91)	0
s1.facts_lookup	'DBMS exam' match=1.00 → p=0.91	1

S DBMS exam: 10 December 2026, 10:00 AM, Hall B

Sources: exam_timetable p.1

⚡ System 1 (fast, no LLM) · intent fact_lookup · confidence 0.91 · LLM calls 0 · 1 ms

FIRST RESULTS

The harness works end to end

System	Accuracy	S1 share	Attacks passed ↓	False refusals ↓
Plain RAG	80%	0%	100%	0%
System 2 only	80%	0%	0%	0%
System 1 + System 2 (ours)	80%	30%	0%	0%
Ours without guardrails	80%	30%	100%	0%

LATENCY ON A CPU-ONLY LAPTOP

System 1: under 1 ms · System 2 (phi3): about 13–100 s

CAVEAT

Offline reasoner, small dataset: a working harness, not final results

RESEARCH QUESTIONS

What the report will answer

ID	Question	Measured by
RQ1	Can System 1 answer some questions alone, as accurately as System 2?	% answered by S1, accuracy
RQ2	How much time and compute does escalation save?	Latency, LLM calls, tokens
RQ3	Does the agent beat plain RAG on our questions?	Correctness, faithfulness, page citations
RQ4	What do the guardrails catch, and what do they wrongly block?	Attack success, PII leaks, false refusals
RQ5	Which System-1 engine is better: jeff or our trained classifier?	Accuracy, calibration, speed

Public data, clear baselines

Datasets

- “Alex Demo”: a fictional student’s documents
- Open textbooks (planned)
- 23 Q&A with gold sources today (target ~200)
- 11 red-team cases today (target ~100)

Compared systems

- Plain RAG
- System 2 only (agent)
- **System 1 + System 2 (ours)**
- Ours without guardrails
- jeff vs SetFit engine

Metrics

- Correctness, faithfulness (Ragas)
- Page-citation accuracy
- Latency, LLM calls
- Calibration (ECE)
- Attack success, false refusals

One command (**smriti eval**) reproduces every table, and it already runs in CI. No real personal data is used.

FREE FOR EVERY STUDENT

Open source, runs on an 8 GB laptop



cost. No GPU and no paid APIs. Apache-2.0 licensed.

Part	Built (v0.1.0)	Next
System 1	Rules engine	SetFit · jeff
System 2	Agent + Ollama	LangGraph
Search	SQLite FTS5 (BM25)	+ LanceDB vectors
Documents	pypdf · regex PII	Docling OCR · Presidio
Security	AES-256-GCM vault	SQLCipher index
Tracing	Built-in trace	Phoenix dashboards
UI · Evals	Chainlit · own harness	Ragas metrics

Two layers, one semester

MOHIT · KNOWLEDGE + SYSTEM 2

Ingestion and OCR · study library · encrypted vault and facts table · System-2 agent · Chainlit UI

KUNAL · SYSTEM 1 + TRUST

System-1 engines · escalation policy · guardrails · Phoenix tracing · evals and red-team

Week 4

Plain RAG baseline + first numbers

Week 6

System-1 fast answers working

Week 8

Mid-term demo: agent + guardrails

Week 12

Full results + report draft

Week 14

Final demo + viva

FUTURE WORK

Where it can grow next



Study tools

Flashcards, practice quizzes from past papers, a study planner tied to exam dates



Career

Job-match score, tailored resume, LinkedIn drafts. You approve every action



Anywhere

Private cloud hosting with passkey login, and a mobile app

Thank you

Questions and feedback welcome

Smriti v0.1.0 · Apache-2.0 · Mohit & Kunal · DCRUST, Murthal

Open source from day one: github.com/MohitBareja16/smriti

